

Introduction

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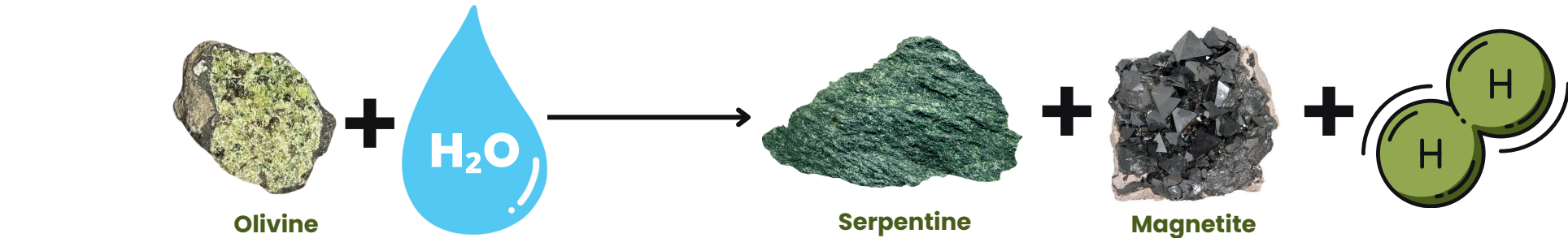
Broader Context:

Hydrogen is a key for decarbonizing sectors like power, transport, and industry. However, current production methods face challenges.

- Grey Hydrogen:** Emits 12 kg CO₂ per kg of hydrogen.
- Blue Hydrogen:** costly CO₂ capturing, transportation, and sequestration.
- Green Hydrogen:** High cost, land-intensive, and dependent on **grid connections**.

Emerging Alternative:

White Hydrogen a **spontaneous, exothermic** reaction between olivine and water produces **low-carbon hydrogen** at **~\$1/kg[1]**.



Figure(1): A schematic figure of Serpentinization Reaction (Created by the finalist using canva 2025)

Current Challenges:

- Poor geological understanding limits field mapping[3].
- Natural olivine has inconsistent Fe²⁺:Mg²⁺ ratios, affecting reactivity[4].
- The serpentinization reaction remains poorly understood and hard to control under field conditions[5].
- Need to explore and locate potential white hydrogen reservoirs for scalable resource development[6].

2.

Research Background:

White hydrogen, also called geological hydrogen, is naturally produced when **ultramafic rocks**, particularly olivine(**forsterite** Mg₂SiO₄ and **fayalite** FeSiO₃), react with **water** under elevated temperatures (100–300°C) and pressures (>500 bar).

This **geochemical process** is known as **serpentinization**.



Thermodynamically, **serpentinization is highly favorable[2]:**

Exothermic ($\Delta H < 0$), releasing heat

Entropy-increasing ($\Delta S > 0$) due to gas formation

Negative Gibbs free energy ($\Delta G < 0$), driving spontaneous reaction.

Field Discovery and Rising Interest:

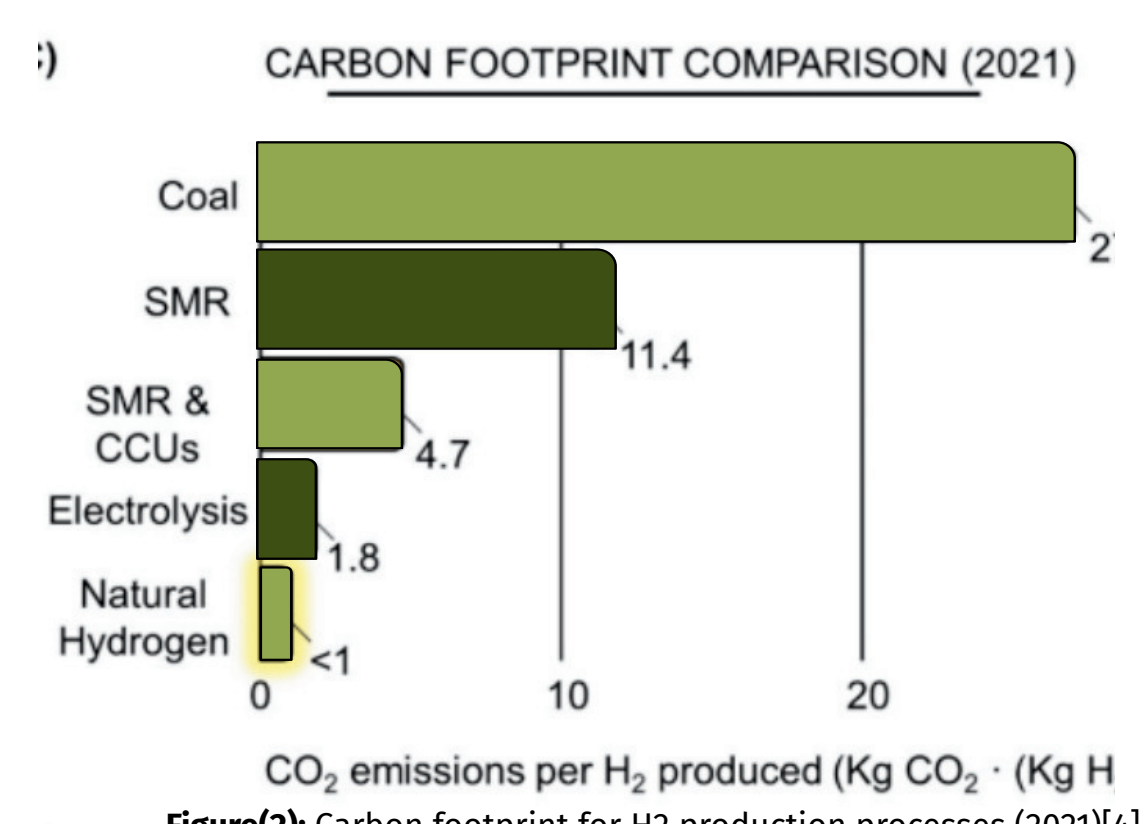
- First field discovery:** (1987) – accidental identification of a natural hydrogen seep.
- Acceleration of scientific focus:** Began around 2010–2012, when white hydrogen's low-carbon potential and economic advantage became recognized
- Previous studies** have explored serpentinization reactions and natural hydrogen seeps, **improving understanding of the geochemical processes involved**.
- However, most research remains limited to **field observations** or natural rock experiments, **without achieving laboratory-controlled replication of serpentinization at scale**.
- This gap motivates the need for **synthetic olivine** with optimized reactivity to enable **predictable and scalable hydrogen generation**.

Cost Comparison:

- Grey Hydrogen:** Emits ~12 kg CO₂ per kg of hydrogen.

- Blue Hydrogen:** costly CO₂ capturing, transportation, and sequestration infrastructure.

- Green Hydrogen:** Carbon-free but remains expensive, land-intensive, and reliant on grid infrastructure



3.

Hypothesis & Objectives

Objectives

- Bridge the knowledge gap** in serpentinization mechanisms by replicating the process using synthetic olivine engineered with **controlled Fe²⁺ reactivity** in a laboratory setting.
- Quantify and compare hydrogen production** between synthetic and natural olivine under identical hydrothermal conditions to evaluate **performance, consistency, and reliability**.
- Assess real-world scalability** by exploring the **integration of synthetic olivine into depleted reservoirs** for large-scale, underground hydrogen **generation and storage with an affordable cost**.

Hypothesis

If **high-purity synthetic olivine** is engineered with **controlled Fe²⁺ reactivity and crystallinity**, and tested under **subsurface-like hydrothermal conditions**, then **white hydrogen** may be produced under **geological conditions with high efficiency**.

4.

Variables

Olivine Synthesis

Independent: Mineral Ratio and reaction conditions

Dependent: Crystalline phase

Control: Calcination Temperature and cooling rate

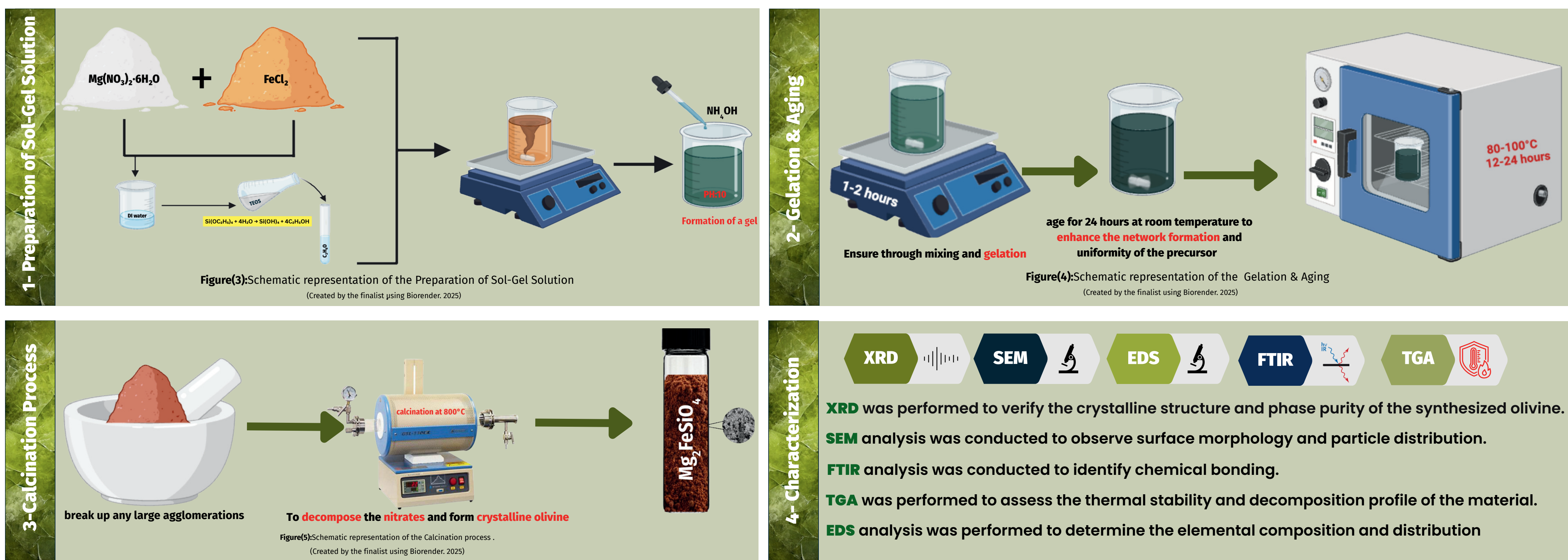
Performance

Independent: Reaction conditions

Dependent: Hydrogen yield

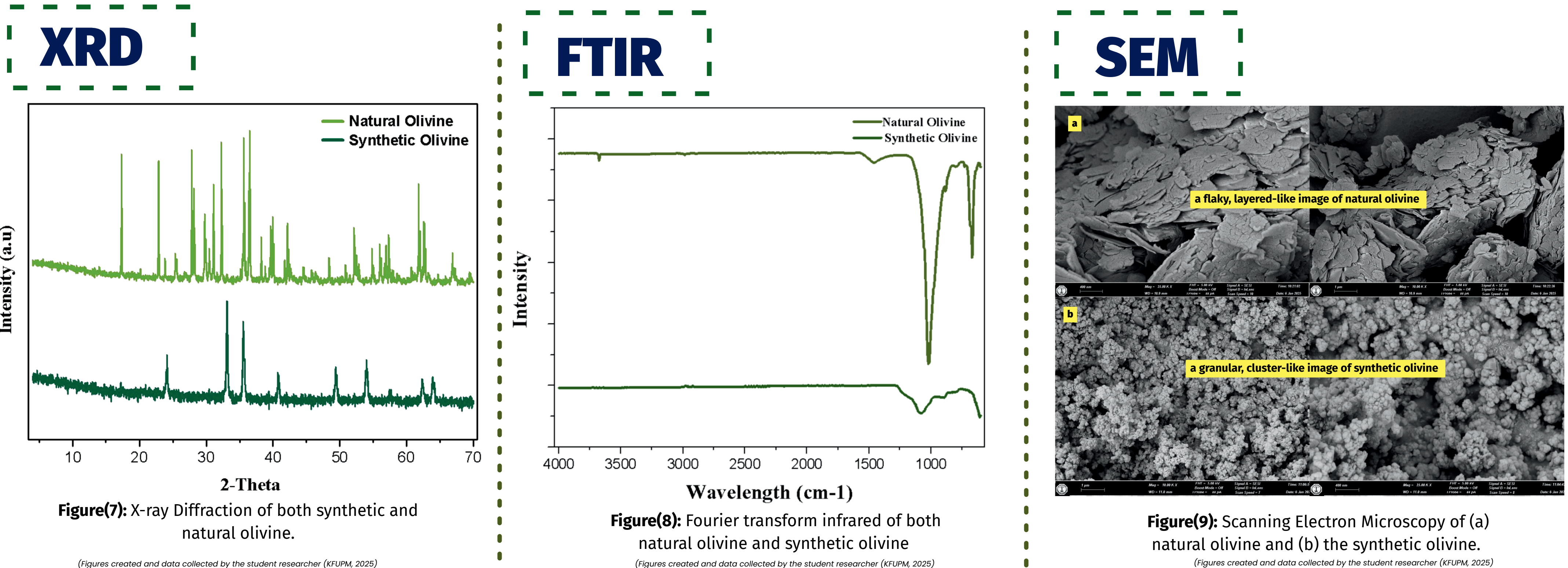
Control: Equipment Set

Methodology

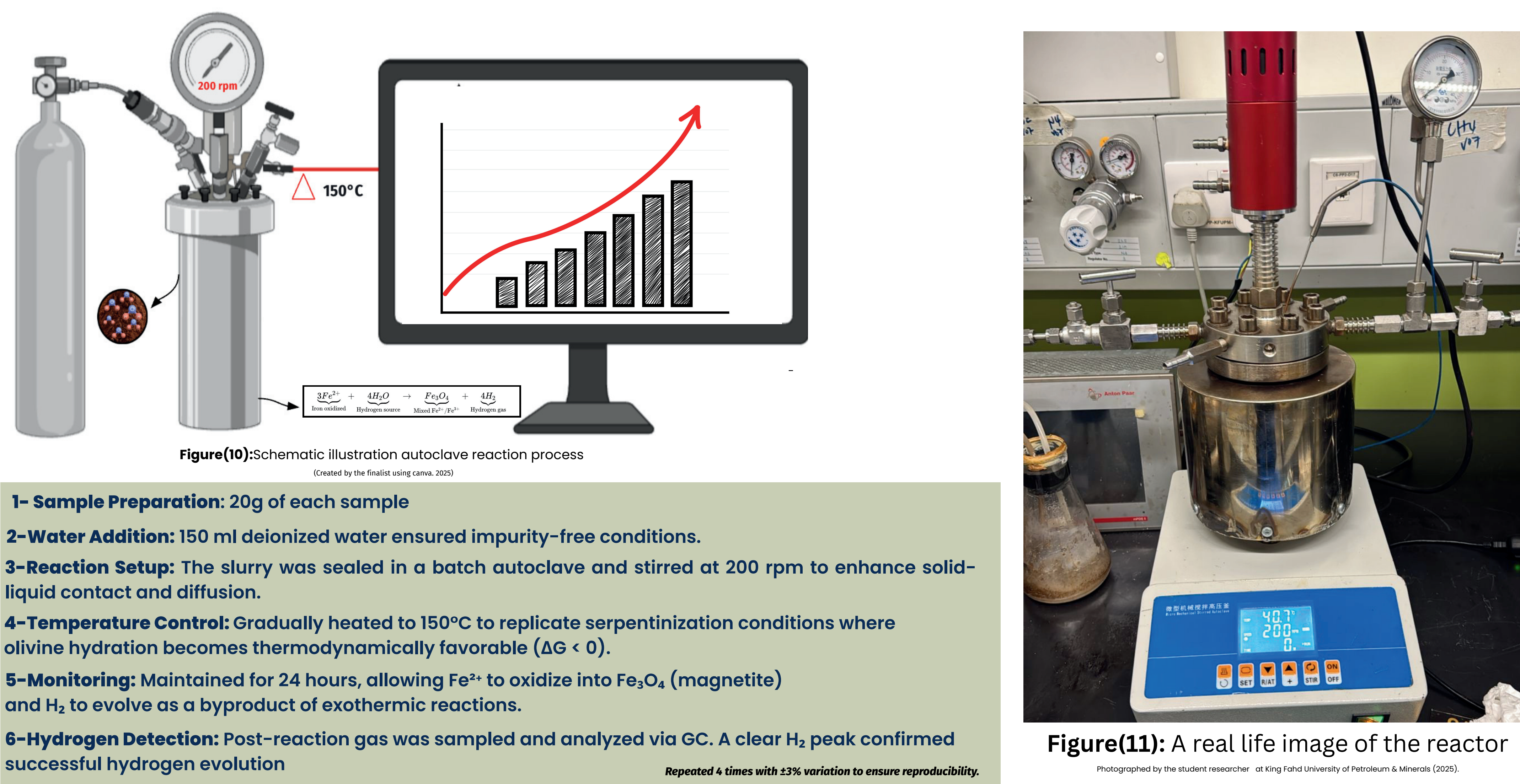


Material Characterization and H₂ Performance

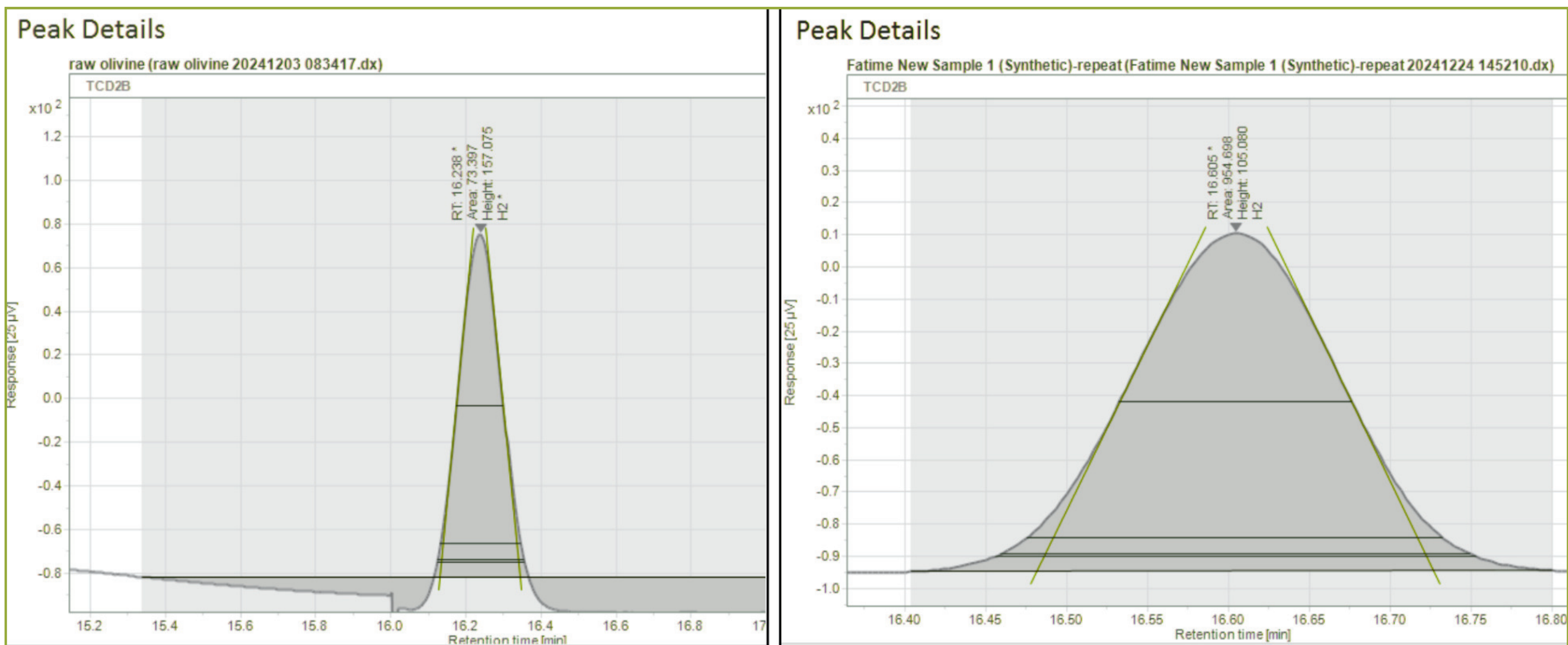
Structural, compositional, and morphological characterization



Simulated Serpentinization Conditions



Performance Evaluation



Figure(12): Gas Chromatography of (a) Natural olivine (b) synthetic olivine

Results

Structural, Compositional, Thermal, and Morphological Characterization:

XRD Analysis (Figure 6):

- >90% crystallinity achieved, with peak positions matching natural olivine and **no secondary phases**, confirming high purity.
- Synthesized olivine exhibited a **single-phase orthorhombic structure**

SEM Analysis (Figure 9):

- natural olivine show **flaky, layered structures**.
- synthetic olivine reveal **granular, cluster-like formations**.

FTIR Analysis (Figure 8):

- key absorption bands corresponding to Si-O stretching and bending at **1000-1100 cm⁻¹**

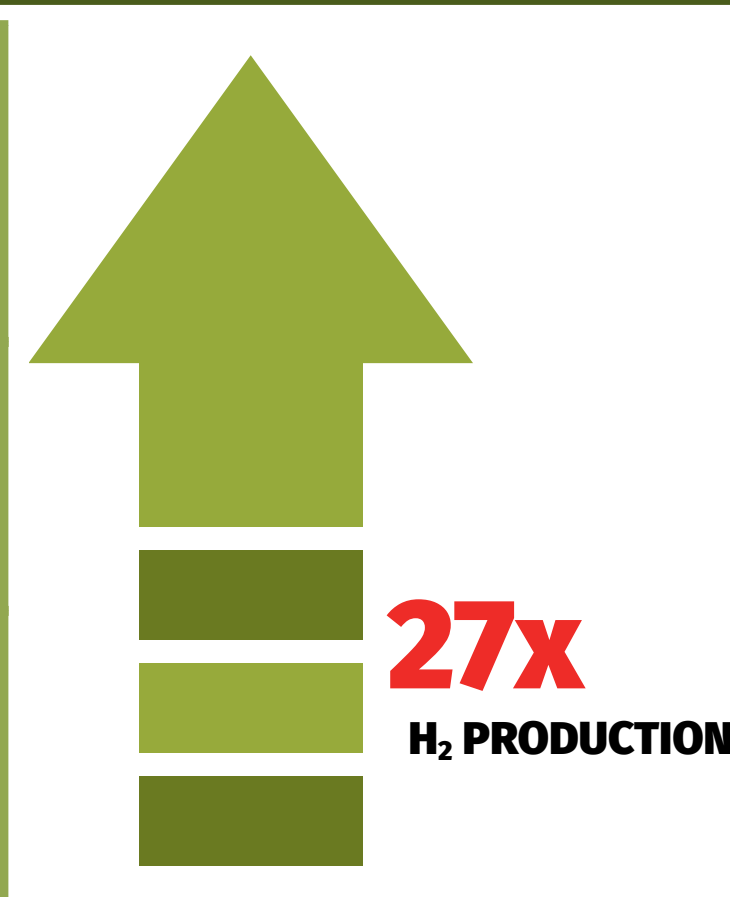
TGA Analysis :

- Synthetic olivine exhibited **lower overall weight loss**, indicating better thermal stability and **less moisture/volatile retention** compared to the natural olivine.

Performance

olivine type	H ₂ production	Reaction control
Natural	150 ppm	Limited by geology
Synthetic	4500 ppm	Engineered for mass production

Repeated 4 times with ±3% variation to ensure reproducibility.



Impact

Process	THIS WORK	Coal	SMR
CO ₂ EMISSIONS	< 1 kg CO ₂ per kg H ₂	~27 kg CO ₂ per kg H ₂	~11.4 kg CO ₂ per kg H ₂
COST PER KG OF H ₂	\$1 per kg H ₂	~\$2 per kg H ₂	\$1.5 per kg H ₂

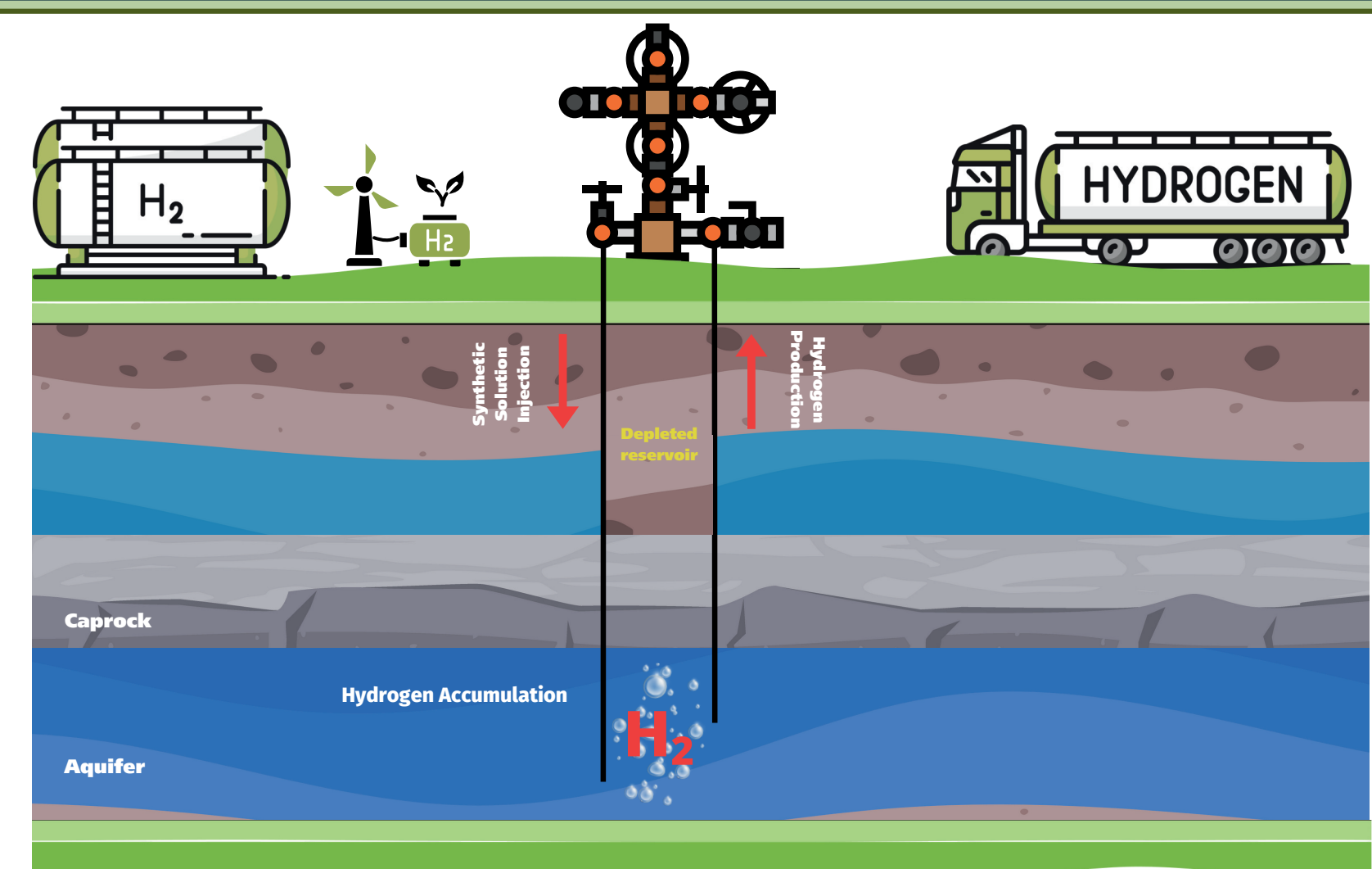
[8] Blay-Roger et al., Earth-Science Reviews, 2023

Conclusion

- Hypothesis proven:** Fe²⁺-rich synthetic olivine produced **27* more hydrogen**
- Advancement:** Offers a **controlled, scalable alternative** to natural serpentinization.
- Reservoirs as solution:** Depleted reservoirs act as both **natural reactors and storage sites**.
- Impact:** A technique that **enables large-scale commercial** white hydrogen production just **under your feet**, potentially supporting **hydrogen hubs, rural energy access, and long-term storage strategies in arid regions**.



Field Implementation



Figure(13): Conceptual setup for subsurface hydrogen production. The method can be implemented in depleted reservoirs or aquifers, enabling in-situ H₂ generation and storage. (Created by the finalist using Canva, 2025)

Future Work

Short-term

- Introduce acidic environment
- Ensure material reusability
- Evaluate performance with saline water
- Perform precise calculations

Long-term

- Conduct long-term stability testing
- Perform reservoir-scale simulation studies
- Continue advance pilot-scale deployment planning in collaboration with an industrial partner.

References

1. Brunet, F. (2019). Hydrothermal production of H₂ and magnetite from steel slags: A geo-inspired approach based on olivine serpentinization. *Frontiers in Earth Science*, 7, 17. [PDF](#)
2. Bassez, M.P. (2017). Anoxic and Oxidation of Fe(II)-silicates and Monosulfides as a source of Fe(II) minerals and H₂. *Origins of Life and Evolution of Biospheres*, 47(1), 107–118. [DOI](#)
3. Greenberger, R.N., et al. (2015). Serpentinization and iron oxidation. *Earth and Planetary Science Letters*, 416, 21–32. [PDF](#)
4. Klein, F., et al. (2013). Compositional controls on hydrogen generation during serpentinization of ultramafic rocks. *Lithos*, 178, 55–69. [PDF](#)
5. Shihuiya, T., et al. (2020). Simulating serpentinization as it could apply to the emergence of life using the JPL hydrothermal reactor. *Astrobiology*. [Link](#)
6. Carnat, M., et al. (2010). Serpentinization and associated hydrogen and methane fluxes at slow spreading ridges. *HAL Archives*. [PDF](#)
7. Osselin, F., et al. (2022). Orange hydrogen is the new green. *Nature Geoscience*, 15, 936–941. [PDF](#)
8. Blay-Roger, R., Bach, W., Bobadilla, L.F., & Ramirez Rem, T. (2023). Natural hydrogen in the energy transition: Fundamentals, promise, and enigmas. *Earth-Science Reviews*, 240, 104345. <https://doi.org/10.1016/j.earscirev.2023.104345>